

Day 29 – AI Screening Conversation Flow Engine

1. Objective

The AI Screening Conversation Flow Engine controls the state-based flow of the candidate screening conversation.

The engine manages conversation states, response handling, retry behaviour, clarification, redirection, and continuation after maximum retries.

The purpose is to provide a structured conversation flow for the existing AI screening workflow.

2. Processing Capabilities

The Day 29 engine supports:

- Conversation state management
- State transition handling
- Candidate response evaluation
- Silence detection handling
- Missing or vague response handling
- Off-topic response handling
- Unknown response handling
- Interrupted response handling
- Retry management
- Question repetition
- Question clarification
- Maximum retry handling
- Continuation with available information
- Structured transition output

3. Conversation Flow

The configured conversation flow contains the following states:

Start | v Asking | v Listening | v Evaluating | v Decision | v Completed

The flow represents the major stages of the AI screening interaction.

4. Conversation States

4.1 Start

The initial state before the screening question is presented.

Next state:

Asking

4.2 Asking

The AI asks the candidate the current screening question.

Next state:

Listening

4.3 Listening

The AI waits for and receives the candidate response.

Next state:

Evaluating

4.4 Evaluating

The candidate response is analyzed using available screening and communication signals.

The evaluation can consider:

- Answer intent
- Relevance
- Completeness
- Communication signals

Next state:

Decision

4.5 Decision

The conversation engine decides how to handle the candidate response.

Possible actions include:

- Continue
- Retry
- Clarify
- Redirect

4.6 Completed

The screening conversation has completed successfully.

There is no next state after completion.

5. Response Handling

The engine supports several configured response types.

5.1 Silence

When no candidate response is received, the engine can retry the question.

Configured action:

Retry

5.2 Missing or Vague Response

When the candidate response does not provide enough information, the engine requests additional detail.

Configured action:

Clarify

5.3 Off-Topic Response

When the response is not related to the current screening question, the engine redirects the candidate back to the question.

Configured action:

Redirect

5.4 Unknown Response

When the response type cannot be identified, the engine uses the configured clarification behaviour.

Configured action:

Clarify

5.5 Understood Response

When the candidate response is understood, the conversation can continue to the next configured state.

Configured action:

Continue

5.6 Interrupted Response

When the response appears to have been interrupted, the engine requests the candidate to continue.

Configured action:

Retry

6. Retry Policy

The conversation flow uses a configurable retry policy.

Maximum retries:

2

The configured retry actions are:

1. Repeat question
2. Clarify question

After the maximum number of retries is reached, the engine uses:

Continue with available information

This allows the screening workflow to continue without repeatedly asking the same question.

7. Transition Handling

The engine provides a structured transition method.

Each transition can contain:

- Current state
- Next state
- Response type
- Action
- Message
- Retry count
- Continuation status

This provides a consistent interface for downstream conversation components.

8. Configuration

The conversation flow is configuration-driven.

Configuration file:

data/conversation_flow_configuration.json

The configuration defines:

- Conversation states
- State transitions
- Response rules
- Response messages
- Retry policy
- Maximum retry count
- Retry actions
- Post-retry behaviour

9. Implementation

Main implementation:

screening_ai/conversation_flow_engine.py

Configuration:

data/conversation_flow_configuration.json

Automated tests:

tests/test_conversation_flow_engine.py

10. Integration

The Conversation Flow Engine provides a control layer for the AI screening interaction.

It can work with existing screening components such as:

- Answer Intent and Understanding Engine
- Screening workflow
- Transcript processing
- Communication signal analysis
- Screening scoring
- AI screening report generation

The engine controls the conversation flow rather than replacing the existing evaluation components.

11. Testing

The Day 29 automated test suite validates:

- Configuration file availability
- Initial state
- State transitions
- Unknown state handling
- Understood response handling
- Silence response handling
- Vague response handling
- Off-topic response handling
- Unknown response handling
- Retry policy
- Second retry behaviour
- Maximum retry behaviour
- Successful transitions
- Silence transitions
- Maximum retry transitions
- Clarification transitions
- Interrupted response handling
- Complete flow output
- Completed state handling

Test Result

All Day 29 Conversation Flow Engine tests passed successfully.

19 passed in 0.11s

12. Day 29 Deliverable

The completed AI Screening Conversation Flow Engine provides a configurable state-based conversation layer for the screening workflow.

It manages:

- Conversation states
- Response handling
- Retry behaviour
- Clarification
- Redirection
- Interrupted responses
- Maximum retry handling
- Structured conversation transitions

The implementation is configuration-driven and supported by automated tests.

The Day 29 test suite completed successfully with 19 passing tests.